

Comparative Analysis of RNN Architectures for WTI Crude Oil Price Forecasting

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Abstract—This report compares three recurrent neural network architectures, SimpleRNN, GRU and LSTM, for one-month-ahead forecasting of the West Texas Intermediate (WTI) crude oil price. The dataset, taken from Kaggle, holds about 19 years of monthly observations ending in December 2024: the WTI price and eight explanatory series (the EUR/USD exchange rate, oil inventories, production, rig counts, inflation, and three indicators derived from the price itself). Each network reads a window of 90 months and is evaluated on the last 28 months with the root mean square error (RMSE) and the mean absolute error (MAE). The LSTM and the GRU reached an RMSE of 5.86 and 5.91 USD per barrel (7.52% and 7.58% of the mean test price) and an MAE of 4.86 and 4.85 USD, while the SimpleRNN reached an RMSE of 10.00 (12.83%) and an MAE of 8.18 USD. The gated architectures were more accurate and trained more stably than the SimpleRNN, and their results were too close to rank against each other. All three models follow the general price level but not sharp month-to-month movements. The report also describes how the models were developed and the limitations of the experimental setup.

Index Terms—crude oil price forecasting, recurrent neural networks, SimpleRNN, GRU, LSTM, time series forecasting, deep learning, economic indicators

I. INTRODUCTION

Time series forecasting plays an important role in financial and economic decision-making, particularly in the energy sector. Forecasts of crude oil prices matter to investors, policy-makers and industry, and the price is hard to predict because it is volatile and depends on many economic factors. Recurrent neural networks are a common choice for sequential data because they can capture temporal dependencies.

This study implements three recurrent architectures, the Simple Recurrent Neural Network (SimpleRNN), the Gated Recurrent Unit (GRU) [1] and the Long Short-Term Memory network (LSTM) [2], and compares how well they forecast the WTI crude oil price from a set of economic indicators and price-derived series. Section II describes the data and the forecasting task, Section III the models and how they were developed, and Section IV the results. Sections V and VI discuss the findings and the limitations of the setup.

II. DATA AND PROBLEM SETUP

A. Dataset

The dataset is the WTI oil price prediction dataset published on Kaggle [3], which combines series from the U.S.

Energy Information Administration (EIA) and Federal Reserve Economic Data (FRED). It holds about 19 years of monthly observations and ends in December 2024.

The target is `wti`, the monthly WTI price in USD per barrel. The eight input series are: `eur_usd` (EUR/USD exchange rate), `inventory` (oil inventory level), `production` (production volume), `rigs` (rig count), `inflation` (inflation rate), `wti_6m_rolling` and `wti_12m_rolling` (6- and 12-month rolling averages of the price) and `wti_6m_lag` (the price lagged by six months).

B. Preprocessing and Windows

All nine columns are scaled to $[0,1]$ with a min-max scaler fitted on the full dataset. The scaled series is cut into overlapping windows of 90 consecutive months. The input of a window is the 90×8 matrix of the eight input series; the target is the WTI price in the month immediately after the window. The task is therefore a one-month-ahead forecast. The price itself is not among the inputs: its history enters only through the two rolling averages and the six-month lag.

The windows are split chronologically, 80% for training and 20% for testing. This gives about 110 training windows (seven mini-batches of 16) and 28 test windows, whose targets are the months from September 2022 to December 2024. In this period the price moved between about 70 and 89 USD, with a mean of about 78 USD.

C. Evaluation

Predictions are transformed back to USD per barrel and compared with the actual prices y_t over the $n = 28$ test months:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_t (y_t - \hat{y}_t)^2}, \quad \text{MAE} = \frac{1}{n} \sum_t |y_t - \hat{y}_t|. \quad (1)$$

Both are also given as a percentage of the mean actual price in the test months.

III. MODELS

A. Architecture

The models are built with the Keras API of TensorFlow [4]. The three final models share one structure: three recurrent layers with 128, 128 and 64 units and tanh activation, a dense

TABLE I
FINAL MODEL CONFIGURATIONS

	SimpleRNN	GRU	LSTM
Recurrent layers	128, 128, 64 units, tanh		
Batch normalisation	after each layer	after each layer	none
Dropout	0.3, 0.3, 0.3	0.3, 0.3, 0.3	0.3, 0.2, 0.2
Optimiser (learning rate)	RMSprop (10^{-3})	RMSprop (10^{-3})	Adam (5×10^{-4})
Parameters (approx.)	67,000	194,000	254,000
Epochs run	33	36	18
Best epoch (restored)	18	21	3

layer with 50 units and ReLU activation, and a single linear output. They differ in the recurrent cell and, for the LSTM, in regularisation and optimiser (Table I).

B. Training

All models are trained with the Huber loss ($\delta = 1$), a batch size of 16 and at most 100 epochs. The test windows are passed to Keras as validation data, and two callbacks monitor the validation loss: *ReduceLROnPlateau* halves the learning rate after five epochs without improvement (minimum 10^{-5}), and *EarlyStopping* ends training after 15 epochs without improvement and restores the weights of the best epoch. No random seed is set.

Because the target is scaled to $[0, 1]$, absolute errors are almost always below $\delta = 1$, so the Huber loss acts as half the squared error in this setup.

C. Model Development

The final configurations were reached in steps, which Table II lists in the order of the notebook. The first SimpleRNN used one layer and a 30-month window. Adding a second layer, dropout and early stopping, then a third layer with a longer window, and finally batch normalisation, the Huber loss, RMSprop and a 90-month window brought the RMSE down to 10.00. The same configuration was then used for the GRU and the LSTM. For the LSTM, removing batch normalisation, switching to Adam with a lower learning rate and reducing the dropout in the later layers lowered the RMSE from 6.92 to 5.86.

Runs 1–4 use shorter windows, so their test periods are longer (40 and 34 months) and include the price peak of 2022. Their errors are therefore not directly comparable with those of runs 5–9.

IV. RESULTS

Table III gives the errors of the three final models on the 28 test months. Figs. 1–6 show their forecasts and loss curves.

A. SimpleRNN

Training stopped after 33 epochs, with the best validation loss at epoch 18. The RMSE is 10.00 (12.83%) and the MAE is 8.18 (10.50%). The forecast (Fig. 1) follows the price in parts

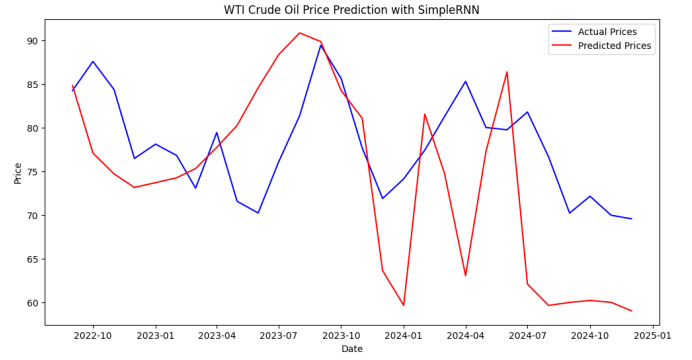


Fig. 1. SimpleRNN: actual and predicted WTI price in the test months.

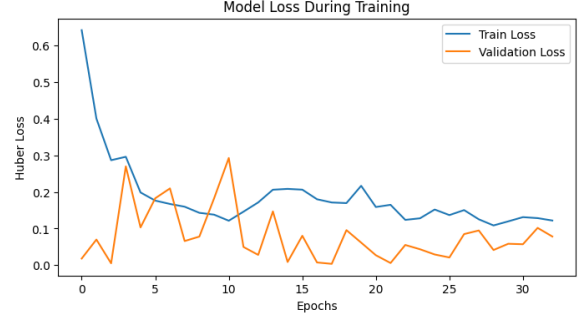


Fig. 2. SimpleRNN: training and validation loss.

of 2023 but moves strongly from month to month, and it falls to about 60 USD around January 2024 and again from July 2024, while the actual price stays between 70 and 82 USD. The validation loss (Fig. 2) fluctuates throughout training.

B. GRU

Training stopped after 36 epochs, with the best validation loss at epoch 21. The RMSE is 5.91 (7.58%) and the MAE is 4.85 (6.22%), a clear improvement on the SimpleRNN. The forecast (Fig. 3) is a smooth curve between about 74 and 81 USD. It follows the general level of the price but not the peaks of September 2023 and April 2024 or the low of June 2023. The loss curves (Fig. 4) are more stable than those of the SimpleRNN.

C. LSTM

Training stopped after 18 epochs. The best validation loss was at epoch 3, so the evaluated weights are those after three epochs of training. The RMSE is 5.86 (7.52%) and the MAE is 4.86 (6.23%), almost the same as for the GRU. The forecast (Fig. 5) is smoother still: it rises from about 76 to 79 USD and then declines to about 73 USD, following the overall trend without responding to individual months. After about the seventh epoch the validation loss lies above the training loss (Fig. 6).

V. DISCUSSION

The gated architectures were clearly more accurate than the SimpleRNN. The GRU and the LSTM reduced the RMSE

TABLE II

ALL RUNS IN THE NOTEBOOK, IN ORDER. RUNS 5, 6 AND 9 ARE THE FINAL MODELS. PERCENTAGES WERE NOT COMPUTED FOR RUNS 1 AND 2

Run	Model	Window	Test months	Main settings	RMSE (%)	MAE (%)
1	SimpleRNN	30	40	1 layer (50 units, ReLU); MSE loss; Adam; 50 epochs, no early stopping	24.24	22.30
2	SimpleRNN	30	40	as run 1 with 2 layers (50, 50), dropout 0.2 and early stopping	12.18	9.51
3	SimpleRNN	30	40	as run 2 (identical code)	22.79 (27.70)	15.64 (19.01)
4	SimpleRNN	60	34	3 layers (128, 128, 64; tanh); dropout 0.3; MSE loss; Adam	18.64 (22.53)	14.49 (17.52)
5	SimpleRNN	90	28	as run 4 with batch normalisation, Huber loss, RMSprop	10.00 (12.83)	8.18 (10.50)
6	GRU	90	28	as run 5	5.91 (7.58)	4.85 (6.22)
7	LSTM	90	28	as run 5	6.92 (8.87)	5.47 (7.02)
8	LSTM	90	28	no batch normalisation; dropout 0.3, 0.3, none; Adam (5×10^{-4})	6.62 (8.50)	5.71 (7.33)
9	LSTM	90	28	as run 8 with dropout 0.3, 0.2, 0.2	5.86 (7.52)	4.86 (6.23)

TABLE III

ERRORS OF THE FINAL MODELS ON THE 28 TEST MONTHS (SEPTEMBER 2022 TO DECEMBER 2024), IN USD PER BARREL AND AS A PERCENTAGE OF THE MEAN TEST PRICE

Model	RMSE	(%)	MAE	(%)
SimpleRNN	10.00	12.83	8.18	10.50
GRU	5.91	7.58	4.85	6.22
LSTM	5.86	7.52	4.86	6.23

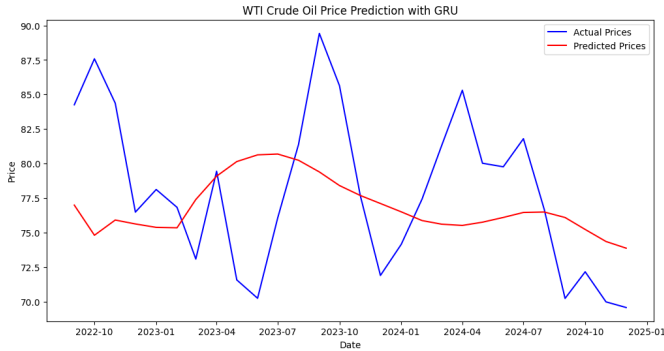


Fig. 3. GRU: actual and predicted WTI price in the test months.

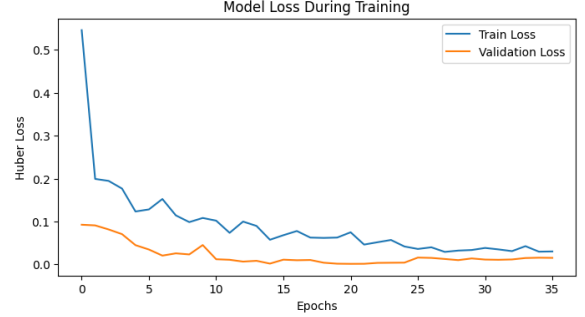


Fig. 4. GRU: training and validation loss.

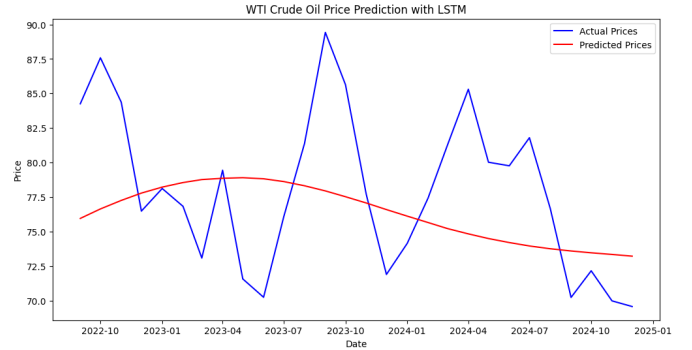


Fig. 5. LSTM: actual and predicted WTI price in the test months.

from 10.00 to about 5.9 USD and the MAE from 8.18 to about 4.9 USD, and their loss curves were smoother. This agrees with the usual expectation that gating helps a network use long input sequences, here 90 months.

The GRU and the LSTM cannot be ranked against each other. Their errors differ by 0.05 USD in RMSE and 0.01 USD in MAE. This is much smaller than the variation between repeated runs seen during development: runs 2 and 3 in Table II use identical code and differ by more than 10 USD in RMSE.

All three models reproduce the general level of the price but not its sharp month-to-month movements. The SimpleRNN forecast is erratic, and the GRU and LSTM forecasts are smooth curves that miss the peaks and lows of the test period. A likely reason lies in the inputs: the price itself is not among them, so the most recent price reaches the networks only through the 6- and 12-month rolling averages, which change slowly.

VI. LIMITATIONS

- *Small sample.* About 110 training windows are used to fit between 67,000 and 254,000 parameters, and consecutive windows share 89 of their 90 months. The test set has 28 months.
- *Validation on the test months.* The test windows serve as validation data for early stopping and learning-rate reduction, and the configurations in Table II were compared on them. The reported errors are therefore likely to be optimistic.
- *Scaling fitted on all data.* The min-max scaler also sees the test period. The effect is probably small, because the test prices lie inside the range of the training prices.
- *No seed and single runs.* Each configuration was run once without a fixed random seed, so results vary between runs

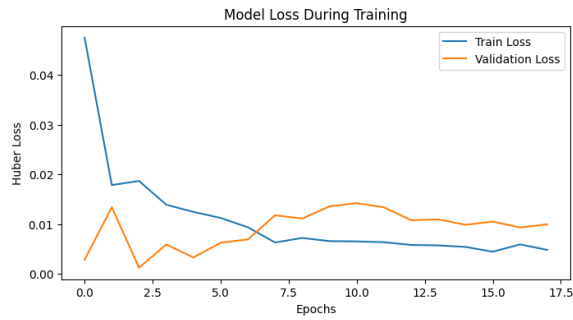


Fig. 6. LSTM: training and validation loss.

and no measure of spread is available.

- *Different settings for the LSTM.* The LSTM uses a different optimiser, learning rate and regularisation from the other two models, so differences cannot be attributed to the cell type alone.

VII. CONCLUSION

Three recurrent architectures were trained to forecast the monthly WTI crude oil price one month ahead from 90-month windows of eight economic and price-derived series. The GRU and the LSTM reached an RMSE of 5.91 and 5.86 USD, against 10.00 for the SimpleRNN, and trained more stably. The two gated models performed equally well within the precision of this experiment. None of the models captured sharp price movements.

Future work could use a separate validation period taken from the training data, with the scaler fitted on the training data only; fixed seeds and several runs per configuration; the same optimiser and regularisation for all cell types; the lagged price as an additional input; a comparison with classical time series models such as ARIMA; and higher-frequency data, which would give more training windows.

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